Gmail Tom Piccolo <tompiccolo50@gmail.com>

STEM and Physics proficiency required to be an animator

4 messages

Tom Piccolo <tompiccolo50@gmail.com> 14 February 2024 at 21:54
To: ian.wilcock@herts.ac.uk, m.bowman@herts.ac.uk, "m.p.bowman@herts.ac.uk" <m.p.bowman@herts.ac.uk>

Hello

I had some questions and am thinking about doing Animation at Herts this year.

I got two A*s and and A at A Level in Biology, Chemistry and Physics. My A was in Physics and an A* in Biology and Chemistry. This was over 2 years ago.

During my A Level studies, I was renowned for my excellent performance across all 3 sciences, consistently on A grades in Year 12 (I got 90% in my Electricity test in Year 12) and 91% in my Year 13 Physics: Paper 1 mock and 84% in my Year 13 Physics: Paper 2 mock (these were A* grades)

I was recognised as gaining above "90% on class tests" in my Year 13 UCAS University statement.

I gained a very high A in my A Level Physics examination in Year 13- 74% or 200/270. This was only 10 marks away from an A*.

In Chemistry, I was in the top 7 percent nationally and the top 4 percent nationally for two of my papers (top 19 percent in the other Chemistry paper).

I later won the award at my sixth form (Bedford Modern School) for the outstanding sixth form scientist.

An A grade in Physics does, however, translate to having a high level of Mathematical proficiency also, in accordance with the PISA OECD assessments, where students are scored based on their performance in the Maths section of PISA and given a score. An A Level Physics grade A would be in Level 5, or advanced proficiency, able to solve challenging questions.

I myself have been able to answer several Level 5 and 6 questions from the Maths section of the OECD PISA assessments, which correlates highly with general intelligence and also academic ability.

I would like to know whether such high proficiency in STEM subjects such as Physics- as demonstrated above is required to be an animator, for instance, working for Dreamworks, Pixar, or Universal in their animated productions, such as How To Train Your Dragon, amongst other things.

Do you need to be excellent or very good at Physics and Maths, or STEM, to be an Animator?

Many thanks

Tom Piccolo

Martin Bowman <m.p.bowman@herts.ac.uk> 14 February 2024 at 23:05
To: Tom Piccolo <tompiccolo50@gmail.com>, "ian.wilcock@herts.ac.uk" <ian.wilcock@herts.ac.uk>, "m.bowman@herts.ac.uk" <m.bowman@herts.ac.uk>

Hello Tom

When you say you are thinking of studying animation at Hertfordshire this year, have you already applied? If you are a UK citizen it is now too late to do so (the deadline was January 31st). If you are an international student then you can continue to apply. I am assuming you are referring to our 3D Animation and Visual Effects degree (we have four animation degrees). Obviously please tell me if you were thinking of one of our other animation degrees.

In regards to your question, no, you do not need special skills in STEM subjects to be an animator – you need art skills for that. There are many other roles in the animation industry where knowledge of STEM subjects is highly useful though – hard and soft body simulation, fluid simulation (including pyro effects and liquids), tools programming, pipeline design, rigging etc – and for all these areas I am referring to the 3D Animation and Visual Effects degree.

Being good at the above areas is a bonus skill for an applicant, especially if the student has already tried some of the above areas in their portfolio and realises the practical use of such theory, but it is not essential. 3D Animation students need to have reasonable maths skills, and those with more advanced maths skills tend to do better in the technical aspects of creating an animation, but if you want to be a character animator that knowledge will not really apply to much of a character's motion. The most important thing we look for is the applicant's art portfolio, which needs to be extremely high quality. I made a video explaining the sort of artwork and showing the quality level required here: https://vimeo.com/841286222

I hope that helps you

Regards

Martin

Martin Bowman
Principal Lecturer & Admissions Tutor
Digital Animation
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https://vimeo.com/uhanimation

From: Tom Piccolo <tompiccolo50@gmail.com>
Sent: Wednesday, February 14, 2024 9:55 PM
To: ian.wilcock@herts.ac.uk; m.bowman@herts.ac.uk; Martin Bowman <m.p.bowman@herts.ac.uk>
Subject: STEM and Physics proficiency required to be an animator

> You don't often get email from tompiccolo50@gmail.com. Learn why this is important

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Tom Piccolo <tompiccolo50@gmail.com> 14 February 2024 at 23:40
To: Martin Bowman <m.p.bowman@herts.ac.uk>

Hello

I mentioned the achievements I had in STEM subjects:

A-A*s in STEM subjects- Biology Chemistry and Physics, at A Level.

Grade 8-9's in these subjects (GCSE Science and Maths) at GCSE

Could you please clarify whether you absolutely need the above achievements and grades in STEM subjects if you are to be successful, at all, the 3D Animation and Visual Effects degree that Herts offers?

Do you seriously need to be in the top quartile of grades (A-A* in Science A Levels) to be anyhow successful in the 3D Animation and Visual Effects degree at Herts?

For insight, could you please advise, based on experience, as to what approximate proportion of successful applicants who gain admission into the 3D Animation and Visual Effects degree at Herts have such high grades (A-A* in STEM subjects)

No numbers or anything- just an approximate rough figure

Many thanks

Tom Piccolo

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Martin Bowman <m.p.bowman@herts.ac.uk> 15 February 2024 at 00:11
To: Tom Piccolo <tompiccolo50@gmail.com>

Hello Tom

"Could you please clarify whether you absolutely need the above achievements and grades in STEM subjects if you are to be successful, at all, the 3D Animation and Visual Effects degree that Herts offers?"

No. You need art skills first of all.

"Do you seriously need to be in the top quartile of grades (A-A* in Science A Levels) to be anyhow successful in the 3D Animation and Visual Effects degree at Herts?"

No. You need to be a strong artist – you need art skills first. STEM skills are just a bonus.

"For insight, could you please advise, based on experience, as to what approximate proportion of successful applicants who gain admission into the 3D Animation and Visual Effects degree at Herts have such high grades (A-A* in STEM subjects) "

Maybe 5 -10%? The majority of our applicants have very high A levels in Fine Art, sometimes Film making, sometimes Graphic Design – those sorts of subjects.

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